

1. What is HTML?

HTML (Hyper Text Markup Language) is used to create and structure web pages.

🔗 It tells the browser:

- what content to display
- how content is organized

🔗 HTML uses **tags (elements)**.

2. Basic HTML Page Structure

```
<!DOCTYPE html>
<html>
  <head>
    <title>My First Page</title>
  </head>
  <body>
    <h1>Hello World</h1>
  </body>
</html>
```

Explanation:

- `<!DOCTYPE html>` → Declares HTML5 document type
- `<html>` → Root element of the page
- `<head>` → Contains metadata (not visible)
- `<title>` → Page title shown in browser tab
- `<body>` → Visible content of webpage
- `<h1>` → Main heading

3. HTML Elements & Tags

Syntax:

```
<tagname>Content</tagname>
```

Example:

```
<p>This is a paragraph</p>
```

Explanation:

- `<p>` → Opening tag
- `</p>` → Closing tag
- Content is inside the tags

Types:

- Container Tag → `<p></p>`
- Empty Tag → `<br>`, `<img>`

4. HTML Attributes

Attributes give extra information to elements.

Syntax:

```
<tag attribute="value">Content</tag>
```

Example:

```
<a href="https://google.com">Visit Google</a>
```

Explanation:

- href → Attribute name
- "https://google.com" → Value
- It tells the link where to go

5. Headings

HTML provides 6 levels:

Example:

```
<h1>Main Heading</h1>  
<h2>Sub Heading</h2>  
<h3>Smaller Heading</h3>
```

Explanation:

- <h1> → Most important heading
- <h6> → Least important

6. Paragraph & Line Break

Example:

```
<p>This is a paragraph</p>  
<br>
```

Explanation:

- <p> → Paragraph
-
 → Break line (no closing tag)

7. Text Formatting Tags

Example:

```
<b>Bold</b>  
<i>Italic</i>  
<u>Underline</u>  
<strong>Important</strong>  
<em>Emphasized</em>  
<mark>Highlight</mark>  
<small>Small text</small>  
<del>Deleted</del>  
<ins>Inserted</ins>
```

Meaning:

- `<b>` → Bold text
- `<i>` → Italic text
- `<strong>` → Important (SEO friendly)
- `<em>` → Emphasized text

8. Links (Anchor Tag)

```
<a href="https://example.com">Visit Website</a>
```

Explanation:

- `<a>` → Anchor tag
- `href` → URL
- Text inside = clickable text

Open in new tab:

```
<a href="https://example.com" target="_blank">Open</a>
```

- `target="_blank"` → Opens in new tab

9. Images

```

```

Explanation:

- `src` → Image path
- `alt` → Alternative text (important for accessibility)
- `width` → Image size

10. Lists

Unordered List:

```
<ul>  
  <li>Apple</li>  
  <li>Mango</li>  
</ul>
```

Ordered List:

```
<ol>  
  <li>First</li>  
  <li>Second</li>  
</ol>
```

Description List:

```
<dl>
  <dt>HTML</dt>
  <dd>Markup Language</dd>
</dl>
```

Explanation:

- → Bullet list
- → Numbered list
- → List item

11. Tables

```
<table border="1">
<tr>
  <th>Name</th>
  <th>Age</th>
</tr>
<tr>
  <td>John</td>
  <td>20</td>
</tr>
</table>
```

Explanation:

- <table> → Table
- <tr> → Table row
- <th> → Header cell
- <td> → Data cell

12. Forms

```
<form>
  Name: <input type="text"><br>
  Password: <input type="password"><br>
  <input type="submit">
</form>
```

Explanation:

- <form> → Form container
- <input> → Input field
- type="text" → Text input
- type="password" → Hidden text
- type="submit" → Submit button

13. Input Types

```
<input type="text">
<input type="email">
<input type="password">
<input type="number">
<input type="date">
<input type="checkbox">
<input type="radio">
```

Meaning:

Each type defines input behavior.

14. Form Elements

```
<label>Name:</label>  
<input type="text">  
  
<textarea></textarea>  
  
<select>  
  <option>Option 1</option>  
</select>
```

Explanation:

- `<label>` → Label text
- `<textarea>` → Multi-line input
- `<select>` → Dropdown
- `<option>` → Option inside dropdown

15. Buttons

```
<button>Click Me</button>
```

Explanation:

Used for actions like submit, click events.

16. Div and Span

```
<div>This is block</div>  
<span>This is inline</span>
```

Explanation:

- `<div>` → Block element (full width)
- `<span>` → Inline element

17. Semantic HTML

```
<header>Header</header>  
<nav>Menu</nav>  
<section>Main Content</section>  
<article>Article</article>  
<footer>Footer</footer>
```

Meaning:

These tags describe the structure of the page clearly.

18. Multimedia

Video:

```
<video controls>  
  <source src="video.mp4">  
</video>
```

Audio:

```
<audio controls>  
  <source src="audio.mp3">  
</audio>
```

Explanation:

- controls → Play/pause buttons

19. Iframe

```
<iframe src="https://example.com"></iframe>
```

Meaning:

Used to embed another webpage.

20. HTML Entities

```
&lt;  
&gt;  
&amp;  
&nbsp;
```

Meaning:

Used for special characters.

21. Comments

```
<!-- This is a comment -->
```

Meaning:

Ignored by browser.

22. Meta Tags

```
<meta charset="UTF-8">  
<meta name="viewport" content="width=device-width, initial-  
scale=1.0">
```

Meaning:

- Defines character encoding
- Makes page responsive

23. Linking CSS

```
<link rel="stylesheet" href="style.css">
```

Meaning:

Connects external CSS file.

24. JavaScript in HTML

```
<script>  
  alert("Hello");  
</script>
```

Meaning:

Adds interactivity.

25. Block vs Inline Elements

Block	Inline
div	span
p	a
h1	img

26. HTML Layout

```
<header></header>  
<nav></nav>  
<main></main>  
<section></section>  
<footer></footer>
```

27. Form Validation

```
<input type="email" required>  
<input type="text" maxlength="10">
```

Meaning:

- required → Must fill
- maxlength → Limit characters

28. Global Attributes

```
<p id="p1" class="text" title="info">Hello</p>
```

Meaning:

- id → Unique identifier
- class → Group styling
- title → Tooltip

29. Accessibility

```
  
<label for="name">Name</label>
```

Meaning:

Helps screen readers and users.

30. HTML5 Features

- Semantic elements
- Audio & video support
- Form validation
- Local storage